Experimental Design & ANOVA

Department of Statistical Sciences, University of Cape Town

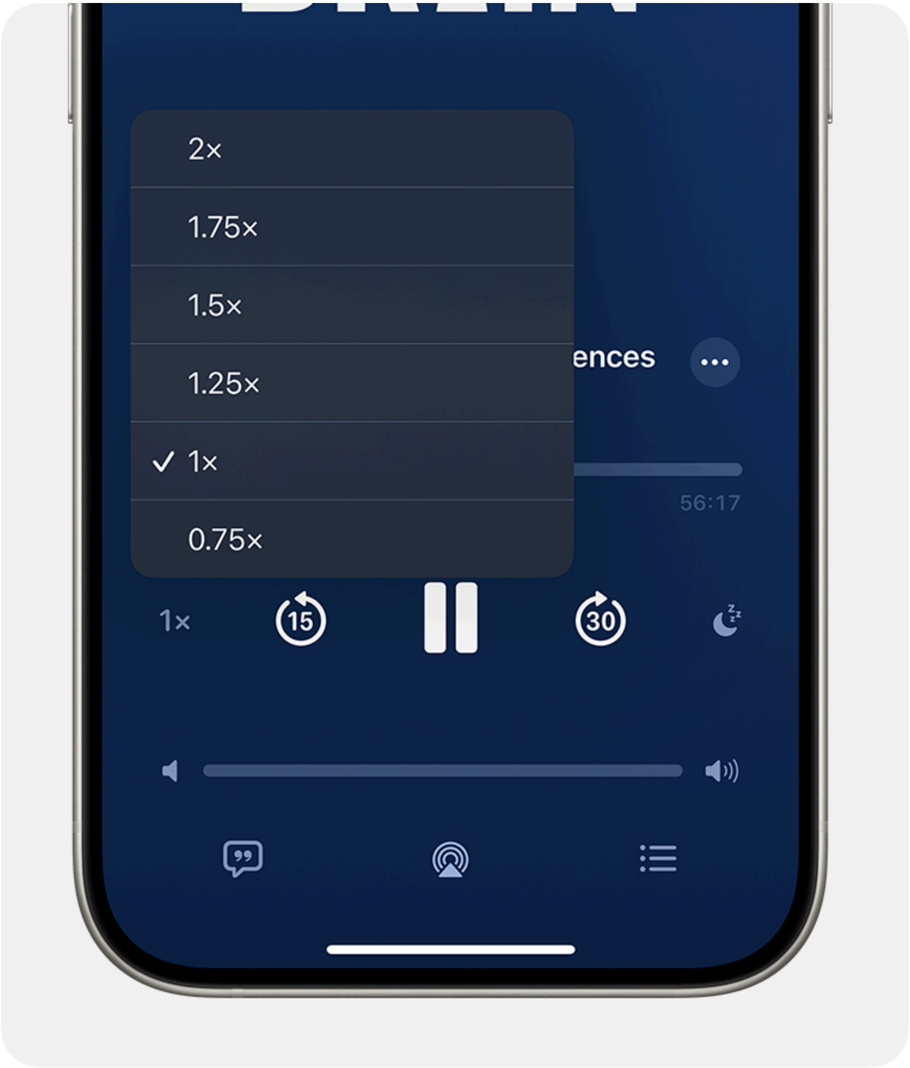
We now move from **regression** to **experimental design & ANOVA**.

- Regression: relating a response to explanatory variable(s), often measured as-is.
- This section: **comparing group means**, did actively changing something (a treatment) change the outcome?

The motivating example

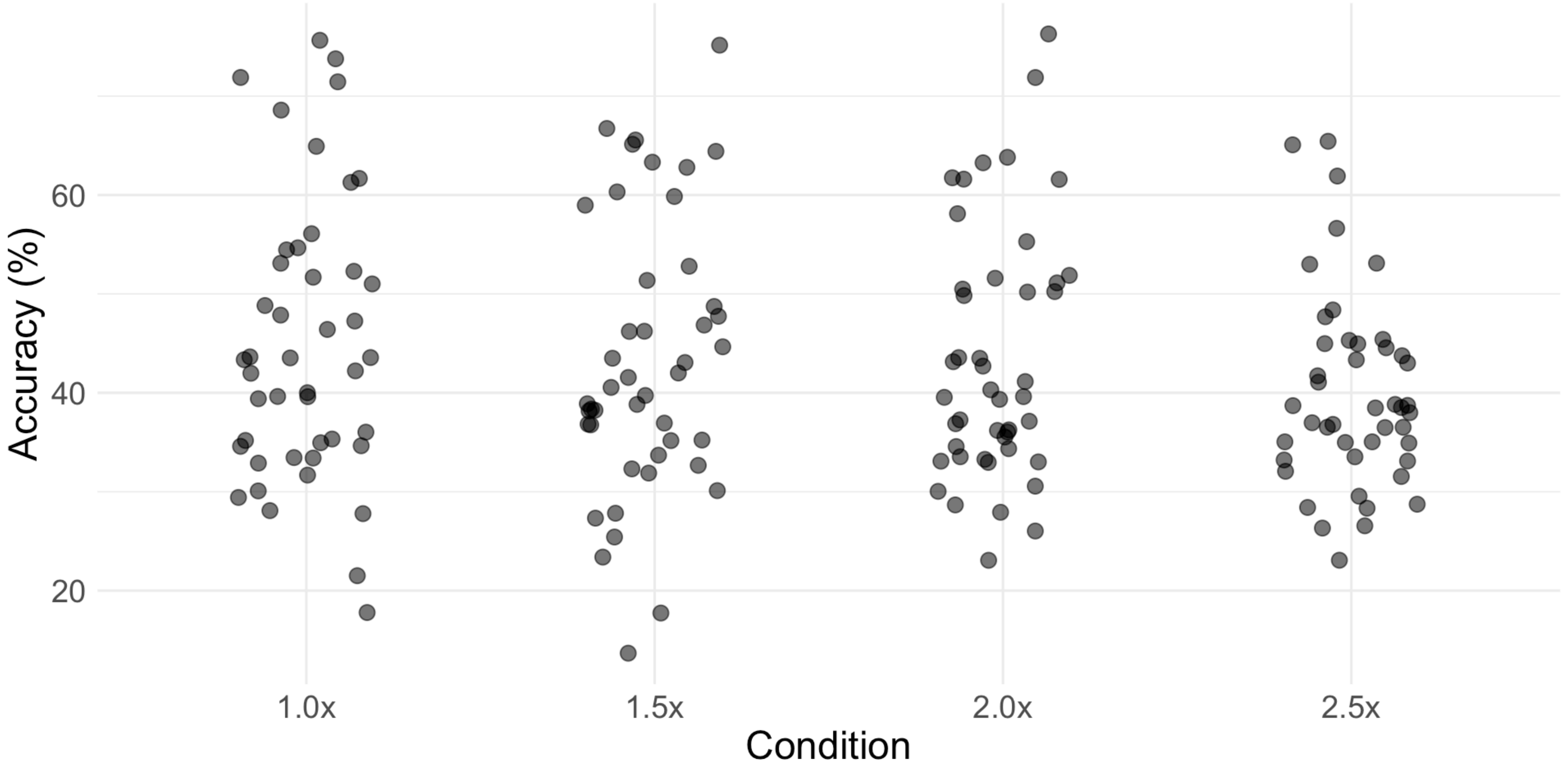
Does **playback speed** affect how much you understand from an audio recording?

A study by Chen et al. (2024) tested this directly.



The motivating example

- 180 participants listened to podcasts at **1x, 1.5x, 2x, or 2.5x** speed.
- Speed was **randomly assigned** to each participant.
- Comprehension was measured with a multiple-choice test.



Where this section is going

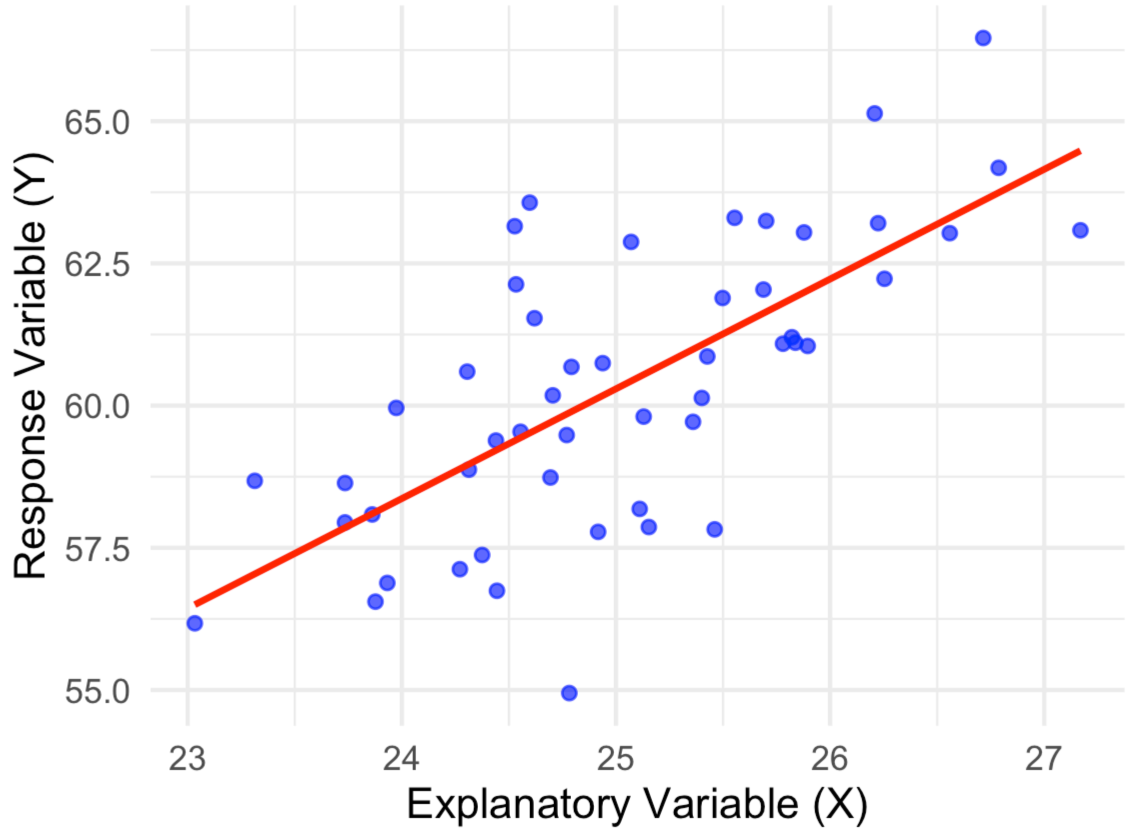
Over the next few weeks:

1. How to **design** a good experiment (randomisation, replication, blocking).
2. How to **build a model** for comparing group means.
3. How to **test** whether group means differ, and check the model is valid.

- Same overall goal as regression: build a model, then use it to answer a question about the world.
- Almost everything you’ve learned about regression **still applies here**.
- ANOVA is not a brand-new idea —> it’s the same idea, dressed differently.

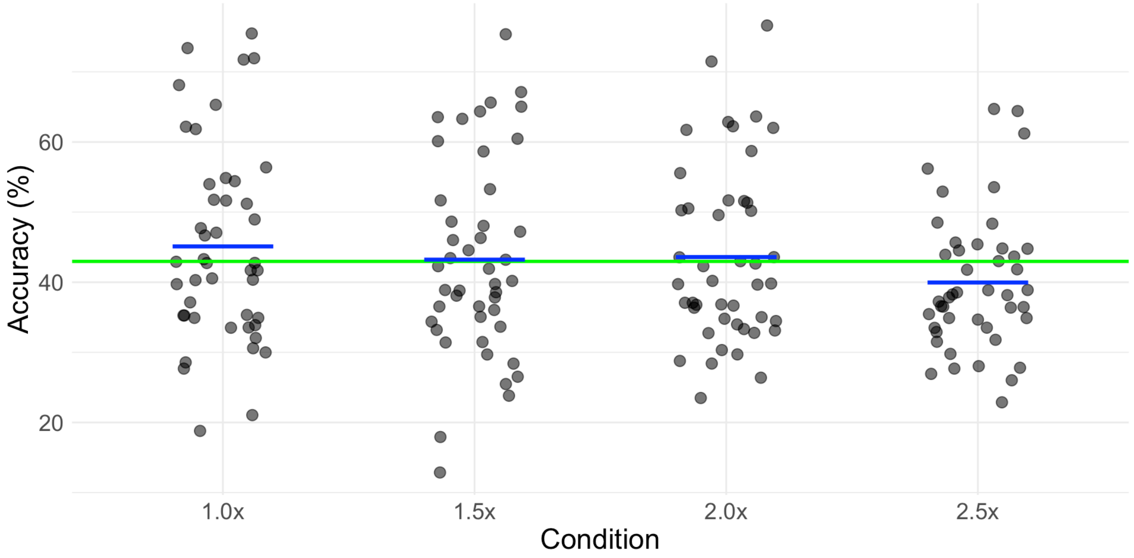
What regression does

- You have a response Y.
- You have an explanatory variable X that you think helps explain or predict Y.
- You build a model that captures the relationship, then use it to explain and predict.



What ANOVA does

- Same response Y.
- Explanatory variable: group membership (e.g. playback speed)
- What *is* new for this section:
- We now also care about **how the data was collected**
- **Treatments from a designed experiment**



The same underlying idea

Both regression and ANOVA:

- Build a model of the form: $\text{response} = \text{explained part} + \text{leftover error}.$
- Estimate that model by making the leftover error as small as possible.
- Ask the same underlying question: **does knowing X tell us anything about Y?**
- No real modelling difference at all!
- So “regression” and “ANOVA” are not two different tools for two different kinds of data

The one idea that carries straight over

In regression, you saw that a model is only useful if it explains **more** of the variation in Y than you'd expect by chance.

- ANOVA asks exactly the same thing, just about *groups* instead of a line:
  - How much does Y vary **between** groups (differences we might care about)?
  - How much does Y vary **within** a group (just noise)?
- If the between-group differences are big relative to the noise, we suspect a real effect.
- This "signal vs. noise" comparison is ANOVA

Wait, isn't ANOVA that table from regression?

- Yes!
- **ANOVA is a procedure**: partition the total variability in Y into an "explained" piece and a "leftover" piece, then compare them.
- In your regression course, you met this procedure as **one part of the output**: the ANOVA table
- In this section, **"ANOVA" is the name of the whole topic**

Same model, different notation

Regression's parametrisation

$$Y_i = \beta_0 + \beta_1 X_i + e_i$$

- Natural when X is numeric, or when you want a prediction equation.
- Works perfectly well on experimental data too — nothing stops you dummy-coding treatments and running `lm()`.

ANOVA's parametrisation

$$Y_{ij} = \mu + A_i + e_{ij}$$

- Natural once you're comparing **group means** rather than predicting along a line.
- Same underlying linear model — just written in terms of a grand mean and group effects instead of an intercept and slopes.

- Neither parametrisation is tied to whether the data is observational or experimental.
- The $\mu + A_i$ form becomes genuinely more useful for convenience later on

Same test

The F-test in regression's ANOVA table and the F-test for comparing group means are **the same formula**:

$$F = \frac{MS_{\text{explained}}}{MS_{\text{error}}} = \frac{SS_{\text{explained}}/df_{\text{explained}}}{SS_{\text{error}}/df_{\text{error}}}$$

- Regression: $df_{\text{explained}}$ = number of predictors in the model.
- Group-mean ANOVA: $df_{\text{explained}}$ = number of groups -1.
- Same ratio, same logic: is the explained variability large relative to the leftover noise?

Literally the same table

Source	df	SS	MS	F
Explained (regression: the line / groups: between-group)	df_1	$SS_{\text{explained}}$	$SS_{\text{explained}}/df_1$	$MS_{\text{explained}}/MS_{\text{error}}$
Error (regression: residual / groups: within-group)	df_2	SS_{error}	SS_{error}/df_2	
Total	$df_1 + df_2$	SS_{total}		

Then why are they taught as different things?

Purely historical!

- **Regression**: Galton and Pearson, late 1800s: continuous variables like height and heredity, mostly observed data.
- **ANOVA**: Ronald Fisher, 1920s, working on crop trials at Rothamsted: comparing yields across experimental treatments.
- Different eras, different disciplines, different notation.

Ask a statistician vs. ask a practitioner

So what’s actually new in this section?

- **Not** a new statistical model, a new parameterisation of a linear model
- Establish casual relationships through experimental design